

Engineering Notebook



30456F

Team Number

MOA

Team Name

Addis Ababa Science and Technology University (AASTU)

School

23/02/2026

Start Date

01/04/2026

End Date

1

Book #

of 1

VEX
ROBOTICS





MOA

30456F

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Team Information



Surafel Lema

Designer /Builder /Electrician
/Driver

1st Year Eng... student at AASTU



Betelhem Mekonnen

Public Relations /Builder
/Strategist/club founder

2nd Year E.M.E. student at AASTU



Yohannes Tadesse

Builder /Tester
/Planner

1st Year Eng... student at AASTU



Dagmawi Abebe

Team leader /Programmer
/Designer

4th Year E.M.E. student at AASTU



Eyoab Shemsu

Builder /Electrician
/Strategist

4th Year E.M.E student at AASTU



Fikrte alemayehu

Social media/strategist
2nd year

Software engineering at AASTU
Wants to study artificial intelligence



Hanna Tesfaye

Resource manager
1st year engineering student at AASTU

Musie Samuel

Builder
2nd Year M.E. student at AASTU

Samrawit gebrie

Social media

*4th Year electrical and computer
engineering student at AASTU*

Mekoyet

builder

*4th Year electrical and computer
engineering student at AASTU*

Abel Alebachew

Resource manager

3er year E.M.E. student at AASTU

Getahun Birhanu

Builder

3^{ed} Year E.M.E. student at AASTU

Mikiyas WeldeSemayat

Social media

*2th Year engineering student at
AASTU*

Team Biography

Team MOA Biography

Team MOA is a robotics team based at Addis Ababa Science and Technology University (AASTU) in Addis Ababa. The team was formed to participate in the VEX V5 Push Back competition, with the main goal of building practical engineering experience through competition. Our team is driven by a shared passion for engineering, especially the excitement of seeing ideas turn into working systems and creating something new from simple components.

The team officially began its journey following a collaboration between AASTU and Ethio Robotics, established through a partnership aimed at developing students with a strong foundation in engineering. On February 23, an introduction session was organized by the founder and CEO of Ethio Robotics to recruit students into the VEX Robotics competition. Although there were initial challenges due to miscommunication, the team eventually connected with mentors who guided us through the basics of robot design, explaining how each component functions and how to effectively use them. As a newly formed team, we started with no prior experience in building robots, but with a strong willingness to learn. This was our first time creating something physical as a team, and through the process, each member began to discover their passion and future interest in engineering. Our mentors supported us in understanding the design process, and we gradually improved through hands-on practice, learning not only how to build but also how to think like engineers. Along the way, we also learned how to work effectively as a team.

Throughout our journey, we faced several challenges. One of the main difficulties was aligning as a team, since members were involved in multiple activities. To solve this, we divided roles and responsibilities more clearly. We also struggled with selecting the right robot design and ensuring structural strength during the building phase. During testing, we experienced issues such as jamming, which affected performance. However, through discussion, trial and error, and continuous improvement, we solved these problems, including reducing jamming by designing and adding aligners.

Despite the challenges, our experience was both successful and enjoyable. We participated in a friendly competition where we achieved second place, which motivated us to improve further. One of the most memorable parts of our journey was the environment we worked in—we often had to sit in uncomfortable positions due to a lack of chairs, but this became a fun and unforgettable part of our experience. We truly enjoyed every moment of the process. Our goals are to win, qualify for higher levels of competition, and continue improving our skills. Team MOA believes in persistence, teamwork, and learning from every experience, and we are excited to grow further in the field of engineering.

Meeting Schedule

Our team followed a flexible meeting schedule based on the different roles and responsibilities within the group. Since our team was divided into programming and design/building groups, each group worked in a way that best suited their tasks.

The programming team mainly held online sessions, allowing them to collaborate remotely, test code, and share ideas efficiently despite time and location limitations. This helped them maintain consistent progress even when physical meetings were not possible.

The design and building team met in person approximately 3 to 5 days per week, focusing on constructing the robot, testing mechanisms, and making physical improvements. In addition, every Wednesday afternoon, all team members were required to meet together to discuss progress, align on decisions, and plan the next steps as a full team.

On average, the team dedicated around 15 to 20 hours per week to the project. Due to limited time and resources, we used free class periods and extended our work into the evenings, sometimes working until 1:00 AM to stay on track and meet our goals.

Through this level of commitment, we were able to make significant progress in a short period of time. At the same time, we ensured balance by using the remaining time for studying and personal responsibilities, so that our academic performance was not affected. This experience helped us develop strong time management skills and teamwork.

February 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday

March 2026

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday

Game Analysis & Strategy Development

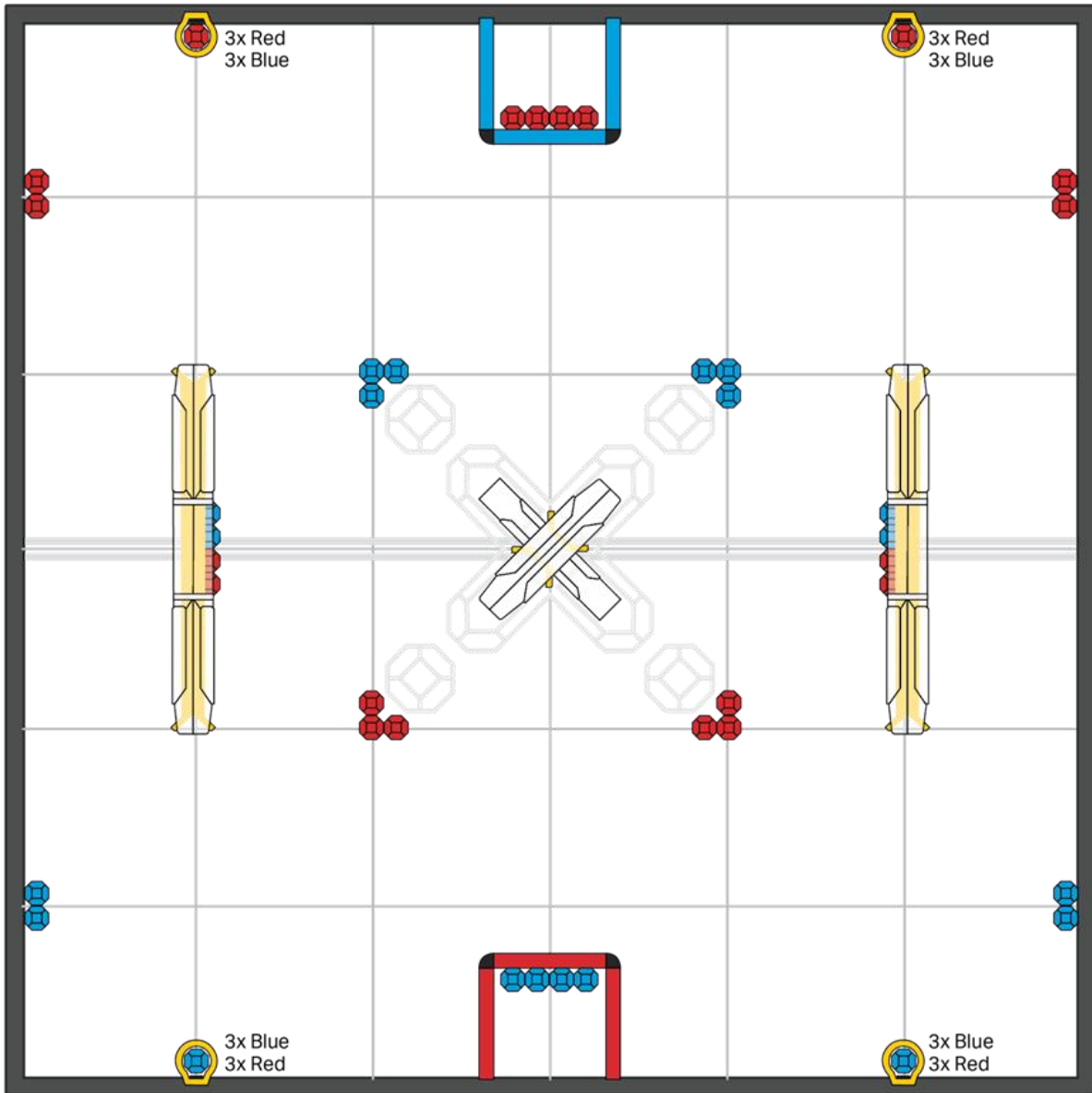
After analyzing the Push Back game, our team decided to use a **balanced strategy** that focuses on scoring, efficiency, and endgame points.

We chose to prioritize **scoring blocks into the top goals**, as they provide consistent points and can be achieved repeatedly during the match. This allows us to build a steady score throughout the game instead of relying only on endgame bonuses. In addition, we focused on creating **controlled bounces**, which help redirect blocks efficiently into scoring areas and reduce time wasted on repositioning.

At the same time, we included **parking** as a key part of our strategy. Since parking provides a high number of points at the end of the match, we planned our gameplay to ensure that the robot is able to reach and secure a parking position before time runs out.

We selected this balanced approach because focusing only on scoring can be risky if time runs out, while focusing only on parking limits total points. By combining both, we ensure that our robot can score consistently during the match and still gain bonus points at the end.

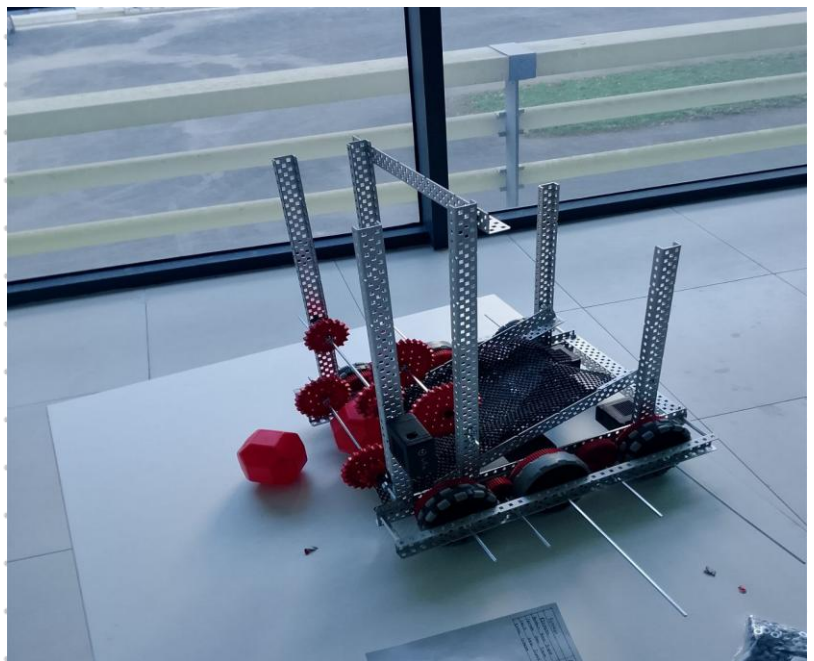
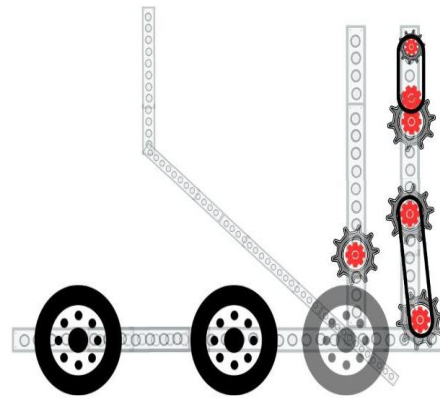
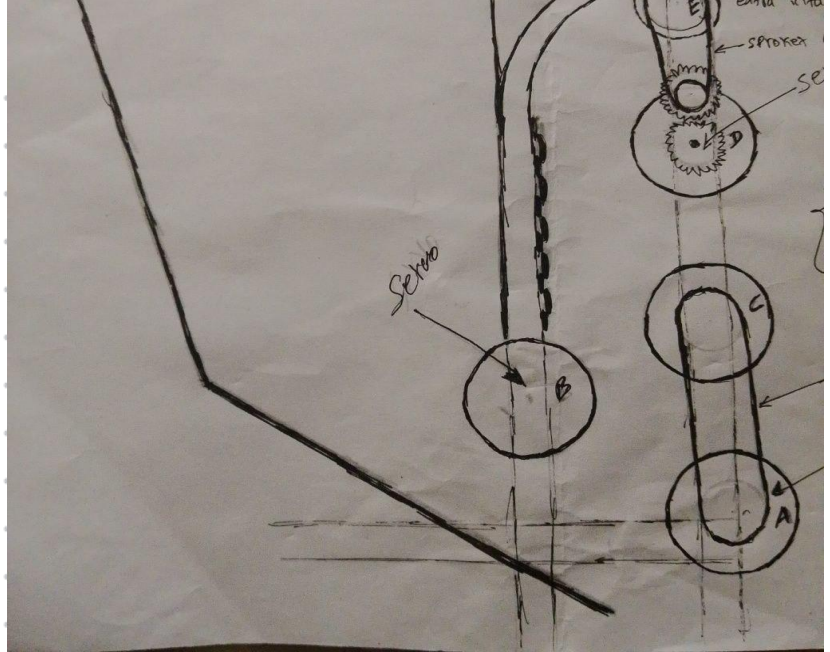
This strategy also influenced our robot design. We needed a robot that is fast enough to collect and score blocks efficiently, stable enough to handle continuous movement, and reliable enough to avoid jamming when interacting with goals. As a result, we focused on building a strong and well-aligned structure that supports smooth scoring and consistent performance.



1/24th Scale

BUILDING PROCESSE

Our design is illustrated in the attached picture, which provides the free hand sketch and side view of the mechanism. From this perspective, you can see that the main components of our system are primarily gear systems and chains. These parts work together to transfer motion and control the movement of the ball. However, we noticed a shortcoming in our current setup: the system is missing one crucial gear. Because of this, the ball cannot reach the middle and top goals, which limits our scoring potential. Additionally, we had not yet decided on the placement of the brain—the main control unit that would coordinate the mechanism—so this is another aspect that requires attention. Despite these issues, we have already begun building the mechanism. Our initial progress allows us to test the basic movement and see how the gears and chains interact, which will help us identify improvements and finalize the design.



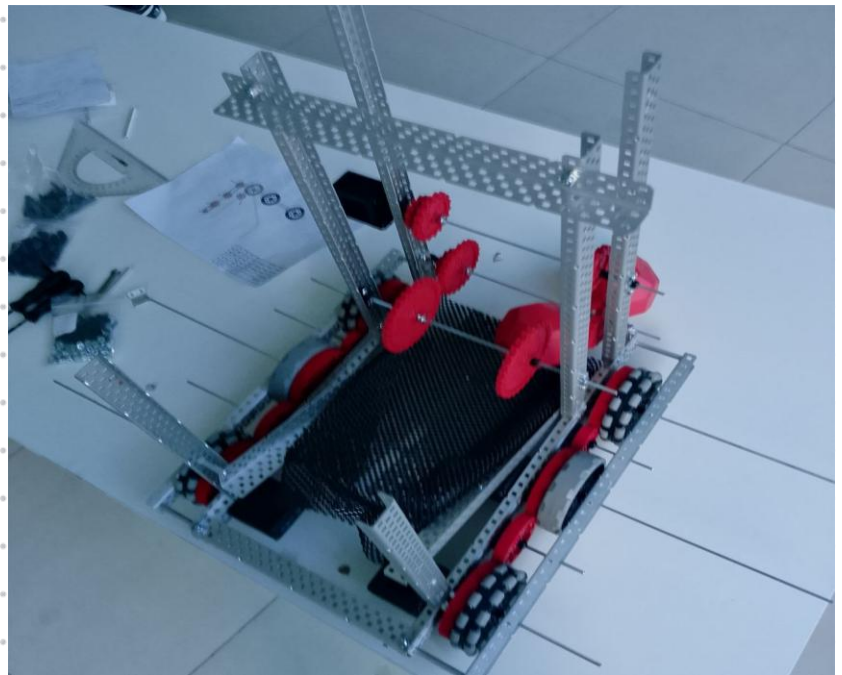
Step 1: Constructing the Base

We started by building a strong and stable base for the robot. The base needed to support all other components, so we used reinforced structural parts to ensure balance and rigidity. The dimensions were checked against our design to make sure the robot would be stable during movement.



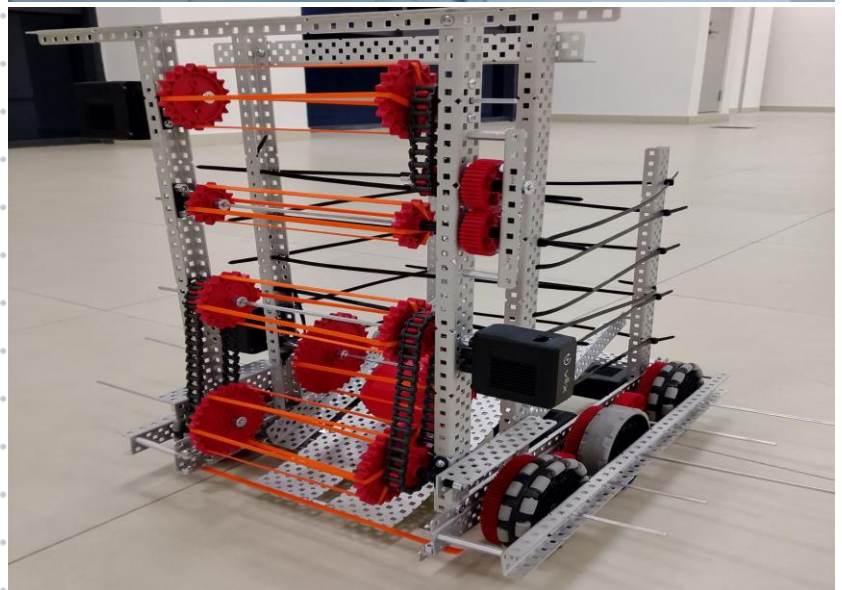
Step 2: Mounting the Gear Systems

Next, we installed the main gears on the base. The gears were aligned carefully to ensure smooth rotation and proper chain movement. At this stage, we realized that one gear was missing, which would later affect the ball reaching higher goals. Despite this, the gear assembly was functional for basic testing.



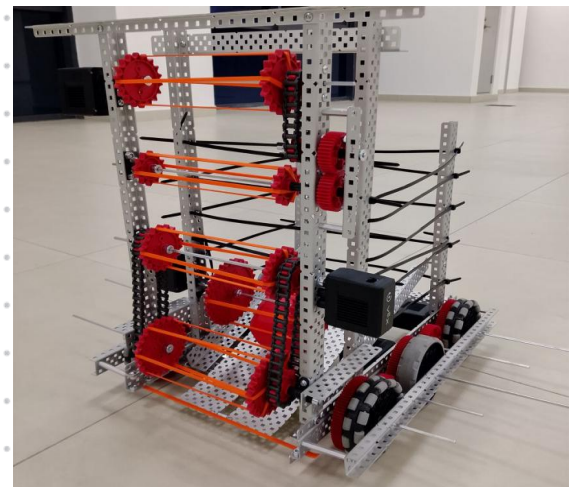
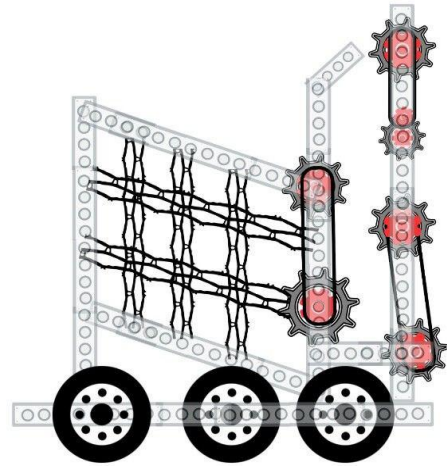
Step 3: Installing Chains

After the gears, we added the chains to connect them. Chains were mounted with the correct tension to prevent slipping and to ensure consistent motion. We tested the gear rotation manually to verify that the chains were transferring motion properly.



Step 5: Updating the Design and Placing the Brain

At this stage, we updated the design to solve problems noticed during the initial build. Adjustments were made to ensure the brain (control unit) and electronics could be placed without interfering with moving parts. The layout was also improved to fix the gear alignment, allowing the ball to reach the middle and top goals, which significantly enhanced scoring potential. After finalizing the updates, the brain and other electronic components were carefully positioned, and wiring was organized to reach all necessary motors and sensors efficiently.



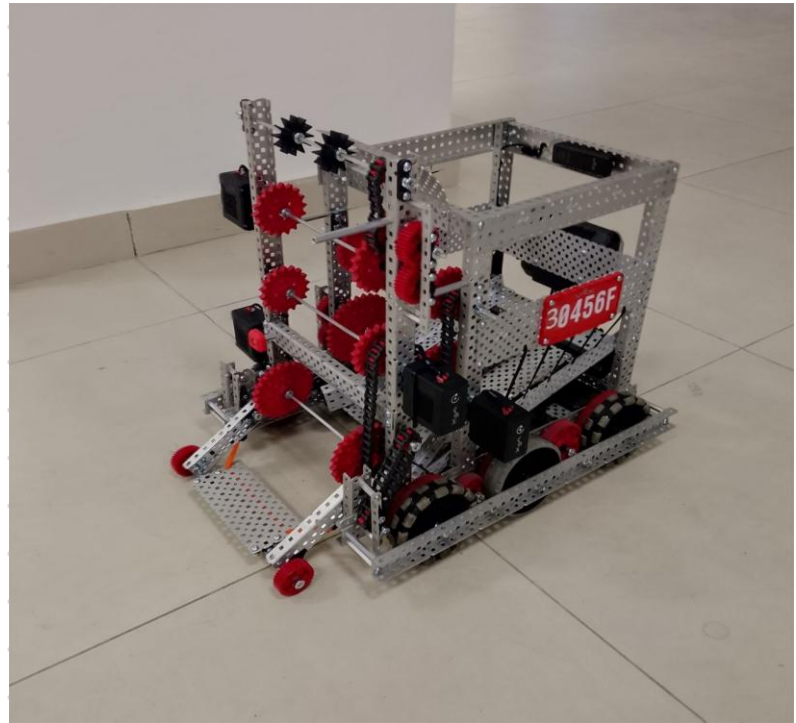
Step 6: Adding Structural Components and Reinforcements

With the updated design and electronics in place, we focused on reinforcing the robot's structure. We added additional structural components to stabilize the mechanism and ensure smooth movement of gears and chains. To improve durability and rigidity, we replaced zip ties with metal supports in key areas, providing stronger connections and preventing wobbling during operation. These structural upgrades also ensured that the robot could handle the motion required to score in the middle and top goals consistently.



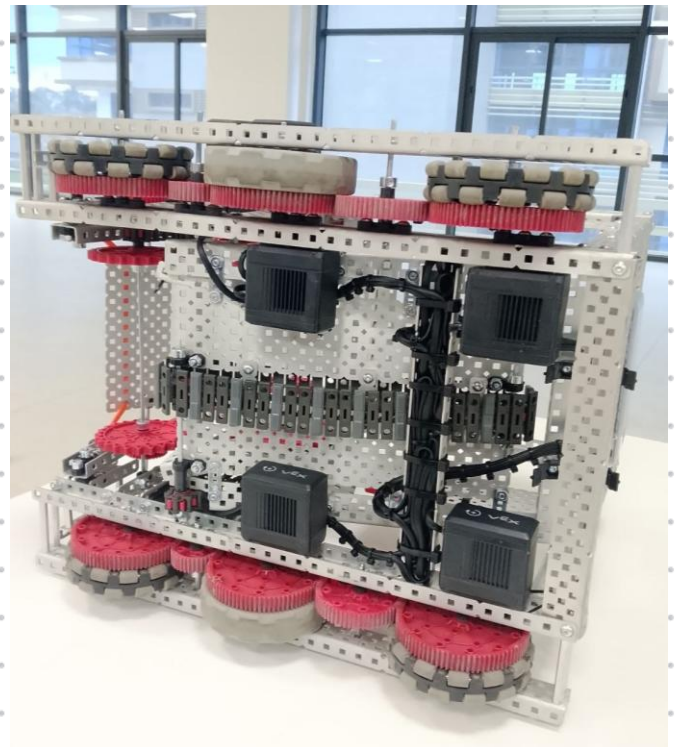
Step 7: Intake Mechanism Installation and Adjustment

In this step, we focused entirely on the intake mechanism. The system was installed and carefully adjusted to efficiently collect balls and deliver them smoothly to the scoring system for the middle and top goals. Proper alignment and movement were ensured to prevent jams or delays during operation. This step was crucial for improving the robot's scoring capability and overall performance.



step 8: Aligners, Cable Management, and Final Adjustments

After the intake mechanism was in place, we added aligners to guarantee precise movement of all gears and moving parts. Cable management was done as a separate process, organizing and securing all wires along the frame with clips or channels to prevent interference with gears, chains, or the intake mechanism. Finally, after multiple trials and adjustments, all components including structure, gears, electronics, and the intake were fully optimized. The robot was ready for reliable performance and thorough testing.



Testing & Results

During the testing phase, we carefully measured and adjusted every component to ensure optimal performance. The robot's movement, gear system, and scoring mechanism were all tested, and no major problems were observed.

- Movement Performance:** The robot moved smoothly, with turns and driving at the intended velocity. Adjustments were made to fine-tune turning and normal speed for better control during operation.

- Scoring Performance:** The ball successfully reached both the middle and top goals consistently. The intake and shooting mechanisms worked reliably, and no jams or misfires were observed.

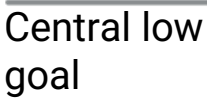
- Gear System:** All gears operated efficiently without slipping or misalignment, ensuring smooth motion and scoring accuracy.

- Observations:** The robot maintained consistent performance across multiple trials, demonstrating that the careful measurements and adjustments made during building were effective.

Overall, the robot performed reliably and accurately in all tested scenarios, confirming that the design, build, and fine-tuning were successful.



Controller mapping



L2

R2

Central top goal

Intake

L1

R1

Long goal



3

24

Null

Null

◀▶4

1◀

Turn

Null



Null

Null



A

Null

Null



Y

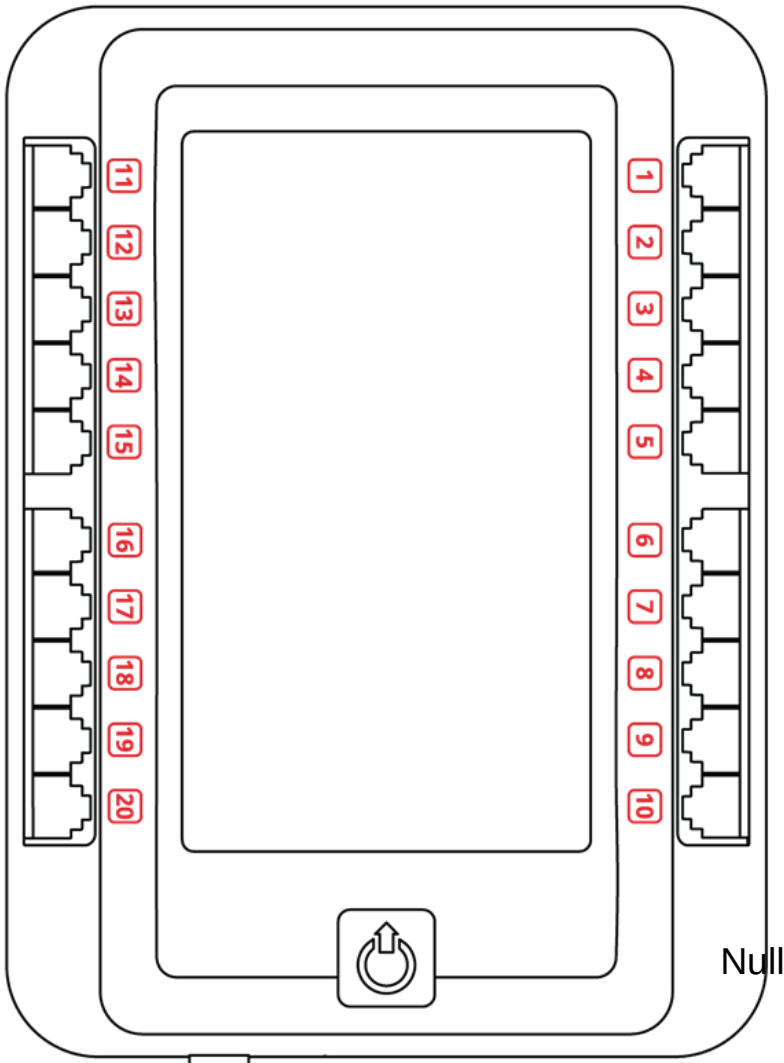
Null

Null

B

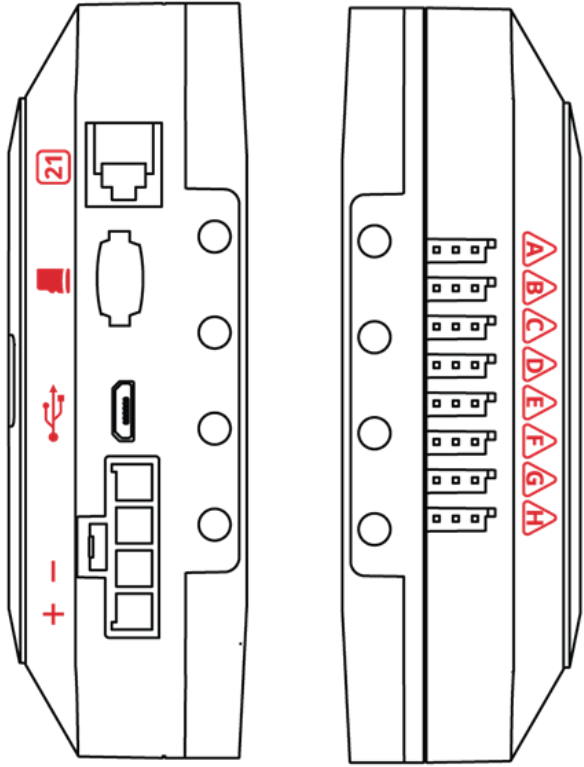
Loader arm

- front left motor 11
- Rare left motor 12
- motor A 13
- Motor B 14
- Null 15
- Null 16
- Motor D 17
- Motor C 18
- Front right motor 19
- Rare right motor 20



- 1 Null
- 2 Null
- 3 Null
- 4 Null
- 5 Null
- 6 Null
- 7 Null
- 8 Null
- 9 Null
- 10 Null

- Radio antenna 21



- A Null
- B Null
- C Null
- D Null
- E Null
- F Null
- G Null
- H Null

Programing

At the beginning of our project, programming the robot was one of the most challenging parts for our team. We had to learn how to control the motors, sensors, and movement using code, which was new to most of us. At first, our robot did not move as expected because of small mistakes in the code, such as incorrect motor directions and timing issues. This caused problems during testing, especially when we tried to score or move precisely. However, instead of giving up, we carefully reviewed our code step by step and tested each function separately.

As we continued improving, we started to understand how to control the robot more accurately. We adjusted the speed, turning angles, and timing to make sure the robot could move smoothly and reach its targets. We also worked on making the controls easier for the driver during the match. By testing multiple times and fixing errors, we were able to improve the performance of our robot significantly. Each mistake helped us learn something new about programming and problem-solving.

In the end, our programming became more reliable and efficient. The robot was able to perform tasks like scoring and moving with better precision. Through this experience, we learned the importance of patience, teamwork, and continuous testing in programming. In the future, we plan to improve our code further by adding more advanced features and making the robot even more autonomous.

Challenges and solutions

During the design and building process, our team faced several mechanical and structural challenges that required careful problem-solving. One of the first issues we encountered was that the ball could not reach the middle and top goals due to a missing or misaligned gear. To address this, we updated the design, added the necessary gear, and carefully adjusted the placement of existing gears. These modifications ensured smooth ball movement and significantly improved scoring performance. Another structural challenge involved the stability of the robot's frame. The original design had weak points that could cause wobbling during operation. To solve this, we reinforced the structure using metal supports instead of zip ties, which provided stronger connections and increased durability under motion. The intake mechanism also presented a significant challenge, as it initially struggled to collect balls efficiently. Through careful alignment, repeated testing, and fine-tuning, we optimized the mechanism so that it could reliably gather and deliver balls to the scoring system. Alongside this, cable management required attention because improperly routed wires could interfere with moving parts. We implemented a detailed cable management system, securing wires along the frame and routing them away from gears, chains, and the intake system. These steps not only prevented interference but also made maintenance easier and improved the overall reliability of the robot. Finally, achieving precise control and consistent performance demanded several rounds of testing and adjustment. We carefully measured turning angles, adjusted normal velocity, and fine-tuned the responsiveness of the robot to the controller. This iterative process ensured that the robot could move smoothly, perform accurate turns, and operate the gears and intake mechanism efficiently. Each challenge we faced—from mechanical alignment to structural stability and electrical organization—provided valuable lessons and opportunities for improvement. By addressing these issues systematically, we were able to create a robot that was stable, efficient, and capable of consistently scoring in both the middle and top goals.

*Conclusion and
future plan*

Conclusion and future plan

In conclusion, the building, testing, and fine-tuning of our robot was a highly rewarding and educational experience. Through careful planning, design updates, and repeated trials, we were able to construct a robot that is mechanically stable, electrically reliable, and capable of consistently scoring in both the middle and top goals. Every stage of the process—from assembling the base and gears to installing the intake mechanism and managing cables—taught us the importance of precision, teamwork, and iterative problem-solving. The testing phase confirmed that the adjustments we made during construction effectively enhanced performance and reliability, demonstrating that careful measurements and attention to detail are critical in robotics.

Looking forward, there are several areas where the robot could be further improved. Future plans include refining the intake mechanism for faster ball collection, exploring more advanced gear and chain configurations to increase efficiency, and integrating sensors for automated scoring. Additionally, enhancing the control system with programmable routines or semi-autonomous functions could improve consistency and reduce human error during competitions. Cable management and structural reinforcements could also be optimized further to make the robot lighter and easier to maintain.

Overall, the project has not only produced a functional and high-performing robot but also provided invaluable experience in engineering design, problem-solving, and team collaboration. By building on this foundation, future iterations of the robot have the potential to perform even more effectively and achieve greater success in competitive settings.